

EMEFA A. APALOO

United States

theCyberSecurityNotebook.substack.com | github.com/emmy-aa

EDUCATION

M.S., Computer Information Sciences(Cybersecurity) · Harrisburg University of Science & Technology (GPA 4.0)

Expected 05/2028

B.A., Computer Science · Minor: Finance · Hanover College

08/2021 – 05/2025

PROFESSIONAL EXPERIENCE

Cybersecurity Analyst / Security Operations Center Analyst Intern · Cybanext Technologies | Remote

09/2026 – Present

- Triage simulated 12-alert client shifts under need-to-know controls, correlating 4 alerts into one critical incident escalated to IR through formal incident-reporting and chain-of-command procedures.
- Close 58% of the queue as non-threats and recommend 3 rule-tuning fixes to reduce alert noise in a regulated client environment.
- Conduct security assessments and adversary-focused analysis across 4 log sources and Splunk (SPL), investigating brute force, macro malware, a port scan, and 2.3 GB of exfiltration. Validate weaknesses via attack-path analysis and map 6 attack stages to MITRE ATT&CK.
- Deliver IR reports with 7 prioritized containment actions, aligned with GRC controls and critical infrastructure defense.
- Threat hunt 60 hosts across 7 days of logs, escalating a missed follow-on compromise and ruling out a false lead. Write behavior-based detection rules that would have alerted about 5 minutes earlier.

Student IT Helpdesk Technician · Hanover College — IT Department | Hanover, IN

08/2023 – 05/2025

- Administered Active Directory user accounts — password resets, MFA configuration, and group membership management — directly supporting enterprise IAM operations
- Logged, triaged, and resolved 30+ monthly Tier 1 incidents via ticketing system; escalated complex issues to senior IT staff with full documentation and clear communication throughout
- Delivered Tier 1 support across hardware, software, and network via in-person, phone, and remote channels for students, faculty, and staff

Website Development & Design Intern · Madison Travel Company | Madison, IN

06/2024 – 08/ 2024

- Diagnosed and resolved backend issues across 20+ plugins — including configuration failures and performance bottlenecks; applied security patches to close known vulnerability vectors
- Increased site traffic by 50% through SEO and content improvements; produced 3 internal style guides and documentation resources adopted for ongoing team use

SECURITY PROJECTS

Home SOC Lab — SIEM Deployment & Network Monitoring. Independent Project.

07/2026

- Built a segmented multi-VM lab using VMware Workstation — Windows 11 endpoint, Ubuntu Server 24.04, and Kali Linux configured as an attack workstation with Nmap, Wireshark, tcpdump, and Netcat, on a host-only network with NAT for controlled web access
- Deployed and configured Wazuh SIEM on Ubuntu Server; installed and validated the Wazuh agent on the Windows 11 endpoint, confirming log ingestion and dashboard visibility
- Diagnosed and resolved infrastructure issues during deployment, including a DNS resolution failure and a downed network interface, through Netplan reconfiguration and resource tuning

Automated HashiCorp Vault Deployment via IaC

10/2025

- Engineered end-to-end infrastructure automation pipeline using Packer and Vagrant to build and provision Ubuntu Server environments, demonstrating hands-on proficiency in IaC principles and secure system configuration
- Deployed and configured HashiCorp Vault for secrets management: implemented secure credential storage, RBAC access control policies, and role-based authentication following industry best practices
- Managed full VM lifecycle including SSH-based remote access and provisioning validation; produced technical documentation for deployment reproducibility

Enterprise Multi-Site Network Infrastructure Design

10/2025

- Architected enterprise-grade network for a 6-location hospitality company (400+ users) with segmented VLANs for PCI-DSS-compliant POS systems, guest Wi-Fi, and employee networks
- Designed redundant infrastructure with AD & RADIUS authentication, QoS traffic prioritization, and automated failover; authored a 3-year TCO analysis report

TECHNICAL SKILLS/ PUBLICATIONS & CERTIFICATIONS

Security: SIEM Triage (Wazuh, Splunk), Log Analysis/Correlation, Threat Detection, Threat Hunting, Detection Engineering, MITRE ATT&CK, NIST CSF, IOC Enrichment, Incident Response/Reporting, Vulnerability Assessment & Management (Nmap), IAM, MFA, HashiCorp Vault, PowerShell Decoding

Infrastructure: Network Security, Cloud Security, TCP/IP, DNS, DHCP, VLANs, QoS, VoIP, Multi-Site Network Design, AWS, Azure, IaC, Packer, Vagrant, Virtualization, Windows / Windows Server, Active Directory, Linux (Ubuntu Server)

Tools & Code: Python, Bash, PowerShell, Claude Code, Claude AI, Java, SQL, JavaScript, Wireshark, Kali Linux, GitHub, Markdown, Microsoft Word/Excel, ChatGPT AI, WordPress, Security & Technical Documentation

Soft Skills: Written Communication, Verbal Communication, Analytical & Critical Thinking, Problem-Solving, Curiosity, Teamwork, Attention to Detail

Publication: "A Rethink of Network Security Through Zero Trust Architecture" — *Contributing chapter author in Examining Vulnerabilities and Adversarial Exploitation of AI and LLMs.*

Certifications: Google Cybersecurity Professional Certificate, CompTIA Security+ (In Progress)